

Drake Olejniczak

AI & Data Engineer · Digital Solutions Specialist

Kalamazoo, MI (Remote / SW Michigan) Drake.Olejniczak@gmail.com (734) 645-5355 github.com/Doogan1 · linkedin.com/in/drake-olejniczak · portfolio.mallard-digital.com/drake-olejniczak

Engineer with a Ph.D. in mathematics and production experience building agentic AI systems, full-stack applications, and data pipelines for local government. Works directly with stakeholders to identify problems and owns solutions end-to-end — from LLM agent loops and vector search backends to React frontends, Cloud Run deployments, and the data pipelines in between. Track record of replacing manual and paper-based workflows with software that real users depend on daily.

TECHNICAL SKILLS

AI / ML	Anthropic Claude API (tool use, vision, batch), OpenAI API (embeddings), RAG pipelines, pgvector semantic search, agentic agent loops, prompt engineering
Languages	Python, TypeScript, JavaScript (vanilla), PHP, SQL
Frameworks	FastAPI, React 18 + Vite, Node/Express, WordPress plugin development, Docker
Data	PostgreSQL + PostGIS, pgvector, Firestore, BigQuery, Cloud Storage, Google Sheets + Apps Script, Power Automate, Looker Studio
Cloud / Infra	Google Cloud Run, Cloud SQL, Firebase Hosting, Cloud KMS, Cloud Build CI/CD, GCP Secret Manager, Cloud Armor, Azure SQL, Docker Compose
Patterns	Agentic AI, RAG, KMS-encrypted PII handling, multi-tenant architecture, OTP/magic-link auth, conflict-aware audit log with undo, async Python (asynpcpg)

EXPERIENCE

Digital Solutions Specialist · Van Buren County — Dept. of Digital Information2024 – Present | Bangor, MI

- Designed and deployed ZIP (zip.dicemi.org), an agentic AI zoning advisor: residents ask plain-English property questions; a Claude tool-use agent performs semantic ordinance search via pgvector and parcel attribute lookups, returning answers with synchronized three-panel UI (map, ordinance viewer, chat). Built solo end-to-end: FastAPI + MapLibre GL JS + PostgreSQL/pgvector + Cloud Run + Cloud Armor load balancer.
- Built a full-stack HUD-compliant grant intake platform (CDBG) replacing paper applications — 10-step guided intake with KMS-encrypted SSN handling (never stored in plaintext), dynamic document checklists, server-side AcroForm PDF packet generation, completeness scoring, and a municipality-scoped staff review dashboard. Stack: React/TypeScript on Firebase Hosting, Node/Express on Cloud Run, Firestore, Cloud KMS, Brevo transactional email.
- Led migration of three SWMPC client websites (swmpc.org, mywaythere.org, us12heritagetrail.com) from legacy Classic ASP / Microsoft Access to WordPress, with hard March 2026 deadline. Engineered the data pipeline: Python + mdbtools extracted relational data from Access .mdb databases into normalized CSVs; PHP/WP-CLI scripts imported into custom post types and 11 custom database tables. Delivered 8 custom plugins and a full redirect map for all legacy .asp URL patterns. Mentored junior developer Maddie through the project.
- Launched stjosephcountymi.gov: scraped, imported, and remapped all content and documents from legacy Granicus platform; deployed county directory integration, events system, and document archive; coordinated DNS cutover with SJC IT director.
- Architected and shipped a modular 8-plugin WordPress ecosystem (county-website-plugins) treating PostgreSQL as canonical data source, with 15-minute automated sync for personnel, departments, and assignments across VBC and SJC; multi-tenant design scoped by county_id. Includes AI-powered WCAG alt text generation via Anthropic Batch API (~\$2/full media library).
- Built a full-stack React 18 + TypeScript / FastAPI staff directory admin (Core DB) with OTP magic-link auth, conflict-aware audit trail (before/after JSON snapshots), and undo system; deployed to Cloud Run with asynpcpg connection pooling.
- Developed and maintain four production AI chatbots across two counties: Marty AI (5,082 conversations / 25,590 messages) and Map Buddy (1,580 conversations / 5,625 messages) for Van Buren County; Joey AI for St. Joseph County. Built custom WordPress Find-a-Ride agent plugin (Claude agentic loop, live at mywaythere.org).

- Drove ADA/WCAG 2.1 compliance initiative across both county websites: Lighthouse/WAVE auditing, NVDA screen reader testing, keyboard navigation testing; improved VBC Lighthouse score from 75 (legacy CityGov theme) to 96 (Blocksy). Coordinated document remediation with all county departments.
- Co-founded Michigan Government AI Network (MI-GAIN), a statewide peer network for government AI practitioners; built public survey dashboard (dicemi.org/migain-survey-and-results) and mi-lgai-repository plugin for member AI resource sharing.
- Established GCP Cloud Run + PostgreSQL/PostGIS as the department's standard cloud-native infrastructure pattern; set up Cloud Build CI/CD pipelines and documented architecture for team reuse across multiple projects.

AI / Data Annotation Consultant · Freelance

2024 | Remote

- Evaluated and refined LLM outputs in mathematics and programming domains to improve accuracy and safety for model fine-tuning pipelines.
- Created structured datasets and reproducible preprocessing workflows supporting fine-tuning at scale.

Assistant Professor, Mathematical Sciences · Purdue University Fort Wayne 2019 – 2024 | Fort Wayne, IN

- Published research in graph theory and network science; applied quantitative and computational methods to complex-system analysis.
- Taught algorithmic thinking, mathematical modeling, and data analysis; mentored student computational research projects in Python.

SELECTED PROJECTS

ZIP — Zoning Intelligence Platform zip.dicemi.org

Python · FastAPI · PostgreSQL + pgvector + PostGIS · Claude API (tool use) · MapLibre GL JS · OpenAI embeddings · Docker · Cloud Run · Cloud Armor

- Agentic AI: user selects a parcel, asks a plain-English zoning question; Claude calls tools (pgvector ordinance search, parcel attribute lookup), synthesizes an answer with panel-control commands, and the three-panel UI synchronizes automatically. Answers are source-cited; system declines rather than fabricates when the ordinance doesn't cover the question.
- Data pipeline: zoning code PDF → section chunks → OpenAI embeddings → pgvector; sub-second relevance-ranked retrieval. Parcel data: St. Joseph County GIS shapefile → coordinate reprojection → PostgreSQL/PostGIS.

CDBG Home Rehabilitation Platform

React 18 + TypeScript · Vite · Firebase Hosting · Node/Express · Cloud Run · Firestore · Cloud KMS · Cloud Storage · Brevo

- Full HUD-compliant federal grant intake: 10-step guided form, SSN encrypted end-to-end via Cloud KMS (never stored in plaintext), dynamic document checklists adapting to each household's composition and income declarations, completeness scoring (0–100), server-side AcroForm PDF packet generation filling all 5 HUD forms with uploaded documents appended.
- Municipality-scoped staff dashboard: status management, document review, applicant messaging, change history audit log. Security architecture validated for PII handling at federal grant program scale.

SWMPC Website Suite swmpc.org

Python · PHP · WordPress · WP-CLI · MySQL · Leaflet.js · ACF Pro · mdbtools · Custom Plugins (×8)

- Migrated three sites from Classic ASP / Microsoft Access to WordPress under a hard deadline. Data pipeline: Python + mdbtools → normalized CSVs → PHP/WP-CLI import into custom post types and 11 custom database tables. Imported 523 meetings, 2,784 traffic count locations, 576 documents (8 GB), 316 points of interest, 56 transit providers across 153 jurisdictions.
- Built custom Leaflet.js interactive map (us12-asset-map) with real-time taxonomy filtering across three dimensions and URL query parameter pre-filtering for region landing pages. Delivered 8 custom plugins including swmpc-permissions (three-tier role architecture across all three sites).

County Website Plugin Ecosystem

PHP 8 · WordPress · PostgreSQL · ACF Pro · WP-CLI · Anthropic Claude Vision + Batch API

- 8-plugin multi-tenant system: County Core (sync orchestration, WP-Cron, REST push endpoints, audit logging), Directory, Menus, Documents (with archive module + expiration notification system), Events (RFC 5545 recurring events + LLM-powered natural-language input), Forms, Alerts, Alt Text (WCAG alt text via Anthropic Batch API, bulk + per-upload hook).

EDUCATION

Ph.D., Mathematics · Western Michigan University
Thesis: [Variations in Ramsey Numbers](#) · Advisor: Ping Zhang
B.S., Mathematics · University of Michigan – Dearborn

2019

2013